

Empirical Proof Record: VALIDATED

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Proof ID: f5ef522e02bf22dc210fd7eda0dad1bb...

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1. Claim

Across a trajectory with spaced re-exposures, the substrate keeps recognising every re-exposed stimulus: (for every same-stimulus re-exposure pair (i, j) in the trajectory, $\text{Jaccard}(\text{concepts_i}, \text{concepts_j}) \geq 0.80$). The reported floor is the worst pair across the whole run.

- **Operationalisation:** Run a schedule of $\text{len}(\text{stimuli})$ unique stimuli each repeated $3\times$ and interleaved ($\text{gap} = \text{len}(\text{stimuli})$) through Kimera's entity target (Takwin). For each stimulus, take the per-cycle "concepts" set at each exposure and compute Jaccard over all exposure pairs. $\text{recognition_jaccard_floor} = \min$ over all pairs across all stimuli; the LTL invariant holds iff $\text{floor} \geq \theta$.
- **Threshold:** $\text{recognition_jaccard_floor} \geq 0.8 \text{ jaccard}$
- **H0:** $\text{H0: floor} < 0.80$ — recognition collapses under manifold deformation somewhere in the run
- **H1:** $\text{H1: floor} \geq 0.80$ — recognition is stable across the whole trajectory (memory-as-deformation holds as a flow property)

2. Verdict

VALIDATED — observed 0.95

recognition floor 0.9500 over 24 re-exposure pairs ($8 \text{ stimuli} \times 3 \text{ exposures}$); mean Jaccard 0.9958; 0 exposures produced no concept set; worst pair: stimulus 2 cycles $2 \leftrightarrow 10 = 0.9500$; control: same-stimulus median 1.000 vs cross-stimulus median 0.026, Mann-Whitney $p=1.7\text{e-}17$ (recognition IS significantly above the cross-stimulus baseline)

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.flow.recognition_stability	recognition_jaccard_floor	0.95	—	ophamin 0.95.0

4. Pre-registration

- Registered at: 2026-05-22T12:58:00.934450+00:00
- Config hash: 0fbe8ce56b2a2db7203910b72aaebc0c...
- Data hash: 4e6090352a40ea27df1cf08fb4d02bd5...

5. Signature

d51ec9783d3af78ad04c021e4ac1f5768b53184f5ff487e85f488248fb0963ad